

AUGUSTINE ISLAND, AK, USA

WMO: 994700

Lat: 59.378N

Lon: 153.348W

Elev: 30

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	5.0	-8.1	3.6	6.5	-4.6	4.3	9.7	56.7	22.6	51.8	22.1	29.4	270	0.894

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	69.1	59.5	66.2	58.3	63.4	56.3	61.7	67.6	59.9	65.7	58.0	63.9	11.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	74.7	64.2	57.0	69.3	62.5	55.1	64.6	59.7	27.4	67.3	26.2	65.8	25.0	64.6	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 48.4	42.7	37.9	DB	-5.0	73.5	5.1	3.7	-8.7	76.1	-11.7	78.3	-14.6	80.4	-18.3	83.0	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	26.1	28.1	28.1	35.4	44.8	50.4	54.9	55.4	49.4	41.4	31.7	28.6
	DBStd	12.75	10.20	7.94	8.33	6.29	5.23	5.08	5.19	4.24	3.94	5.16	8.66	9.19
	HDD50	4243	741	612	678	438	180	52	7	1	54	267	549	664
	HDD65	9281	1206	1032	1143	888	626	441	318	300	468	731	999	1129
	CDD50	446	0	0	0	1	19	63	159	168	36	0	0	0
	CDD65	8	0	0	0	0	0	2	4	2	0	0	0	0
	CDH74	5	0	0	0	0	0	1	4	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.1	20.2	19.9	18.3	15.1	11.6	10.3	11.7	12.4	15.3	17.6	19.5	21.2
Precipitation	PrecAvg	52.9	5.1	4.4	3.4	3.6	3.5	2.9	3.5	4.0	5.8	5.9	5.1	5.7
	PrecMax	63.9	10.2	7.7	6.2	5.7	6.1	4.4	5.1	6.3	10.3	10.6	11.2	9.3
	PrecMin	39.6	1.7	1.9	0.7	1.3	1.3	0.9	1.7	1.3	2.3	2.8	1.3	2.4
	PrecStd	5.6	2.0	1.5	1.4	1.1	1.3	0.9	0.9	1.3	1.7	1.9	2.4	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.9	41.5	42.6	54.9	67.6	72.5	74.0	70.5	60.3	51.3	46.5	44.0
		MCWB	37.7	40.2	40.8	44.1	57.5	64.1	60.2	61.1	53.3	46.6	44.3	42.7
	2%	DB	40.2	39.6	40.4	49.4	60.0	66.7	69.7	67.8	57.6	49.6	45.1	42.1
		MCWB	38.3	38.1	38.9	41.5	48.4	58.4	58.6	59.9	51.2	46.6	43.8	41.0
	5%	DB	38.8	38.0	39.0	45.6	55.8	62.5	66.7	64.9	55.6	48.4	43.7	40.5
		MCWB	37.2	36.7	37.4	39.0	45.5	55.6	58.1	57.7	50.4	45.7	42.4	39.3
	10%	DB	37.2	36.7	37.4	42.1	52.6	58.4	63.6	62.3	54.1	47.4	42.3	38.8
		MCWB	35.6	35.3	35.3	38.3	44.6	52.9	56.5	55.6	50.1	44.9	40.9	37.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	41.4	42.5	46.0	54.7	64.4	63.3	63.7	56.1	50.1	46.1	43.7
		MCDB	40.6	41.5	43.5	56.6	65.9	70.2	70.3	68.3	58.1	50.4	46.2	43.9
	2%	WB	39.2	39.7	40.8	43.2	49.0	60.4	61.4	61.5	54.1	48.0	44.6	41.8
		MCDB	39.6	40.5	41.1	49.9	55.4	63.9	68.4	67.3	55.2	48.7	45.0	42.3
	5%	WB	37.7	38.0	39.1	41.4	47.5	57.4	59.7	59.8	52.7	46.9	43.4	40.1
		MCDB	38.3	38.8	39.2	44.8	52.8	62.0	66.8	65.3	53.7	47.8	44.0	40.8
	10%	WB	36.2	36.4	36.7	40.0	46.3	54.8	58.1	57.6	51.5	45.8	42.0	38.1
		MCDB	37.3	37.2	37.5	43.0	50.6	58.7	64.9	62.8	52.7	46.7	42.8	39.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	6.5	6.7	6.1	6.7	7.4	7.3	6.4	5.2	5.1	6.5	6.8
		MCDBR	6.0	6.1	6.0	9.2	12.2	13.8	13.2	10.5	7.1	5.0	5.4	6.9
		MCWBR	5.1	5.3	5.0	4.6	6.6	7.9	6.9	5.6	5.3	4.5	4.7	6.0
	5% WB	MCDBR	5.9	5.5	5.0	7.0	10.7	13.2	13.2	10.6	5.7	4.5	4.8	6.2
		MCWBR	5.2	4.9	4.6	4.5	6.3	8.7	7.3	6.4	4.6	4.6	4.8	6.0
Clear-Sky Solar Irradiance	taub		0.257	0.306	0.319	0.354	0.374	0.366	0.373	0.361	0.338	0.315	0.277	0.270
	taud		2.251	2.315	2.335	2.309	2.321	2.430	2.455	2.481	2.510	2.414	2.175	2.022
	Ebn at Noon		179	225	260	269	271	275	270	265	253	220	158	110
	Edh at Noon		13	21	28	34	36	33	31	29	24	19	14	11
All-Sky Solar Radiation	RadAvg		92	242	656	1081	1536	1713	1500	1198	783	371	137	58
	RadStd		13	44	120	162	175	166	150	162	114	42	25	7

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+1.44	+3.89	+3.74	N/A	N/A	N/A	-405	-517	N/A	N/A

Nomenclature: See separate page